

CONNECTING ROBOTIC AND HUMAN HANDS

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MOTIVATION



Figure: How generative AI imagines our problem

- ▶ Industry is dangerous!
- ▶ Alas! — if only there were something that would enable us to carry out perilous tasks just as we would with our bare hands, yet without touching the dangerous stuff...

OUR IDEA

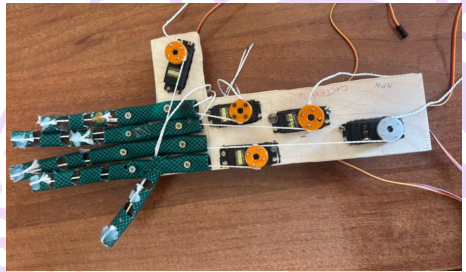
- To build an interface between the human hand and a custom robotic hand through webcams

SEGMENTATION OF THE PROJECT

► System α



► System ω



- ▶ The image processing segment
- ▶ Comprises a Thinkpad and two webcams on a special stand
- ▶ The webcams have matching y axes and orthogonal x axes, enabling production of images suitable for finger triangulation
- ▶ Tasks it takes care of:
 1. camera calibration
 2. image segmentation (finger detection)
 3. finger triangulation
 4. control number inference
 5. message encoding and transmission
- ▶ A special mathematical model of finger movement was conceived so as to satisfy the needs of System α — the 'Kinderica' Model

- ▶ The actuator segment
- ▶ Consists of the robotic hand model, servo motors and an Arduino Uno
- ▶ The hand is fashioned from plywood, garden hose pieces and thin synthetic rope
- ▶ The design leverages the elasticity of rubber
- ▶ Finger movement regulation is achieved through servo motors

RESULTS

- ▶ A functional hand model
- ▶ Successful implementation of numerical control
- ▶ Successful setup of the interface between the subsystems
- ▶ Successful finger triangulation (by colour)

TO DO (tasks of critical priority):

- ▶ Control number inference
- ▶ Implementation of the 'Kinderica' Model
- ▶ Implementation of ML techniques so as to phase out segmentation by colour